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* PL / CC Assignment - 2 *

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* **Title :** Installation and configuration of Google App engine.

* **Problem Statement :** The objective of this lab is to intall and configure Google App Engine (GAE) and deploy a simple web application on it. Google App Engine is a PaaS that allows devs to build and deploy applications in the cloud without managing the underlying infrastructure.

* **Objective :**
  - To understand the concept of Google App Engine (GAE)
  - To install and configure Google Cloud SDK
  - To create and deploy a simple application using Google App Engine.
  - To understand how GAE handles app deployment and scaling.

* **Hardware Requirements :**

A PC/Laptop with 4GB RAM and Internet Connection.

* Software Requirements:

- OS : Any (Windows / Linux / Mac)
- Google Cloud SDK : gcloud command-line tool
- Programming Language:
  - Python (for python-based app)
  - Node.js (for JS-based app)
- IDE/Text Editor : Vs Code or any Text editor
- Web Browser : Google Chrome or any modern one.

* Theory:

1) Google App Engine (GAE)
- It is a Platform as a service (Paas) provided by Google Cloud that allows devs to build, deploy and scale applications without managing servers.

2) PaaS:
- It provides a complete development and deployment environment in the cloud. Developers focus on writing code while the platform manages infrastructure, scaling and maintenence.

3) Google Cloud SDK:
- It is a set of tools used to manage resources and applications hosted on Google Cloud platform using the command line.

4) app.yaml file:
- It is a configuration file used by Google App Engine. It defines the runtime environment, entry point and other application settings

5) Automatic Scaling:
- GAE automatically scales the application based on incoming traffic by increasing or decreasing the no. of instances

* Algorithm / Procedure:

1) Open: https://www.skills.google/ and log in using a Google Account

2) Claim the 35 free credits available on the platform

3) Start the lab titled "Host a web App on Google Cloud using Compute Engine"

4) Do not complete the lab, instead use the temporary Google account credentials provided by the lab to log in. This account is used as a workaround to avoid Google's billing account requirement.

5) Open Google Cloud console in a new tab using the lab credentials.

6) Navigate to API's & Services and search for "App Engine Admin API". Check whether it is enabled; if not, enable it.

7) Install Google Cloud SDK on the system and initialize it using the command:

```
> gcloud init
```

Log in using the lab-provided account and select the available project.

8) Create the App Engine Application by running the command:

```
> gcloud app create
```

Select the appropriate regions when prompted.

9) Open VS Code, create new folder and create the following 3 files:

- main.py : Any simple python application

- app.yaml : Configuration file with the following content:

runtime : python314
entrypoint : gunicorn -b :$PORT main:app

- requirements.txt : Dependency file containing :
Flask
gunicorn

10) Open the project folder in the terminal

11) Run the following commands to deploy and access the application:

> gCloud init
> gcloud app deploy
> gcloud app browse

12) After successful deployment, access the application using the URL opened in the browser.

* Conclusion:

In this lab, GAE was successfully configured and a simple web application was deployed. The experiment helped in understanding cloud deployment and how GAE manages application hosting and scaling automatically.